AI-DRIVEN ATTACK CHAIN PREDICTION AND INTELLIGENT SECURITY RECOMMENDATION SYSTEM

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

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ABSTRACT

The proposed AI-Driven Attack Chain Prediction and Security Recommendation System is an intelligent cybersecurity framework developed to detect, analyse, and respond to network-based cyber attacks. The system uses network flow data from the CICIDS2017 dataset, which contains both benign and malicious traffic representing different attack categories such as DDoS, DoS, PortScan, Bot, Web Attacks, Infiltration, and Brute Force. The collected network traffic is preprocessed by handling invalid and infinite values and preparing the data for machine learning. Random Forest is used as the machine learning classifier to identify and classify different types of network traffic. The performance of the trained model is evaluated using accuracy, precision, recall, F1-score, classification report, and confusion matrix. The complete system further analyses the detected attacks using the MITRE ATT&CK framework to identify the corresponding tactics and techniques. The identified attack stages are used for attack-chain prediction, where a data-driven Markov Chain model analyses the transitions between attack stages and predicts the possible next stage of an ongoing attack. The predicted attack stage is considered along with the detected attack and threat information for dynamic risk assessment. Based on the detected attack, predicted attack progression, and calculated risk level, the system generates appropriate security recommendations to support security analysis and incident response. By integrating Random Forest-based intrusion detection, MITRE ATT&CK mapping, attack-chain prediction, dynamic risk assessment, and security recommendation into a unified framework, the proposed system provides a comprehensive approach to network threat detection, attack analysis, and security decision support.

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CHAPTER 1
INTRODUCTION

1.1 BACKGROUND

Due to the quick expansion of digital networks, cloud computing, web applications, and networked devices, there is a greater possibility of cyberattacks. There is huge volume of network traffic constantly generated and exchanged by organizations which cannot be manually monitored. Cyberattacks such as Distributed Denial of Service (DDoS), Denial of Service (DoS), Port Scanning, Brute Force, Botnets, Web Attacks, and Infiltration can cause confidentiality, integrity, and availability problems of network resources. In traditional IDS, it is necessary to define certain rules for detecting attacks on network resources. Nonetheless, there are some drawbacks of using rule-based IDSs for network traffic because of the huge amount of data that has to be checked. For this reason, machine learning techniques can be applied.

Random Forest algorithm is used to classify network intrusion and attacks using the proposed system. Data pre-processing is done on the network traffic data of the CICIDS2017 dataset and relevant features are fed to the model. The model categorises the network traffic into benign and various attacks. To give a clearer idea about the attacks detected by the system, MITRE ATT&CK framework is included, which classifies the adversary behavior into tactics and techniques. The detected attack is mapped to the MITRE ATT&CK framework. Security guidance for the attack is generated by the system based on the above information.

1.2 PROBLEM STATEMENT

With the growing amount of network traffic and the constant increase in cyberattacks, it becomes quite challenging to manually track and analyze such activities. In addition, traditional IDS usually rely on predetermined signatures and rules, which may fail to recognize the evolving cyberattacks. Machine learning algorithms are able to increase the efficiency of attack recognition based on the recognition of patterns in the network traffic. At the same time, the classification of attack alone cannot give all the necessary information about the attacker and the security countermeasures needed. As a result, there is a necessity to create a system that will automatically classify attacks in the network with machine learning, correlate attacks with the corresponding MITRE ATT&CK tactics and techniques, and offer proper security recommendations. The proposed approach meets such a requirement with the use of Random Forest and MITRE ATT&CK.

1.2 OBJECTIVES

The major goals of the Network Intrusion Detection and Security Recommendation System are as follows:

- To gather and preprocess the data related to network traffic using the CICIDS2017 dataset.
- To extract network-flow features that may be used in the process of network attacks detection.
- To build an Random Forest-based classification model that can classify network traffic as benign or malicious.
- To classify different types of network attacks such as DDoS, DoS, Port Scanning, Brute Force, Botnet, Web Attacks, and Infiltration.
- To map the detected attacks according to MITRE ATT&CK tactics and techniques in order to understand the behavior of the attackers.
- To provide recommendations on the possible actions that should be taken during the process of responding to the detected attack.
- To test the effectiveness of the developed model with the help of accuracy, precision, recall, F1-score, and confusion matrix.

1.4 SCOPE

The scope of Network Intrusion Detection and Security Recommendation System involves detection and analysis of network-based cyberattacks by using machine learning.

- Network Traffic Analysis: The system analyses the flow of network traffic to distinguish between normal and malicious activities.
- Dataset-Based Detection: CICIDS2017 dataset is utilized, which contains both benign network traffic and various attacks.
- Data Preprocessing: Data preprocessing is done for network traffic data by cleaning the network traffic data and preparation of relevant features.
- Machine Learning Classification: Random Forest classifier is used for classifying network traffic into benign and various attack categories.
- Attack Types: Attack identification involves attacks such as DDoS, DoS, Port Scanning, Brute Force, Botnet, Web Attacks, and Infiltration.
- MITRE ATT&CK: Identified attacks are correlated to the MITRE ATT&CK framework in order to find relevant tactics and techniques of the attack.

- Security Recommendation: The system gives security recommendations according to the attack.
- Evaluation of Performance: The performance of the model is evaluated through accuracy, precision, recall, F1-Score, and confusion matrix.

1.5 SECURITY REQUIREMENTS

The above-discussed architecture must ensure the security during the entire process of data processing, training of the model, attacks detection, and recommendations creation. The main security requirements include:

- Data confidentiality: Traffic and security data should be protected from any unauthorized access.
- Data integrity: Datasets and predictions should be protected from unauthorized alteration.
- Authentication and authorization: Only authorized users should have access to the system and its security-related data.
- Secure data processing: The input data should be validated and processed correctly to avoid negative influence on the system from incorrect or malicious data.
- Model security: Trained Random Forest model should be protected from any unauthorized alterations and use.
- Logging and monitoring: Main activities of the system, predictions, and security incidents should be logged.
- Availability: The system should be available to detect and analyze network threats.
- Secure recommendations: Security recommendations should be provided based on the correct classification of attacks and threats to avoid inappropriate actions.

1.6 SDG ALIGNMENT

Table 1.1 SDG Mapping

S. No.	SDG No. & Goal	Relevant SDG Target	Project Contribution / Alignment
1	SDG 9 – Industry, Innovation and Infrastructure	Target 9.1: Develop reliable, sustainable and resilient infrastructure.	The proposed system improves the security and resilience of digital network infrastructure by detecting and classifying cyberattacks using Random Forest.
2	SDG 16 – Peace, Justice and Strong Institutions	Target16.6: Develop effective, accountable and transparent institutions.	The system supports secure digital operations by providing automated threat detection, MITRE ATT&CK-based threat context, and security recommendations for responding to network threats.

CHAPTER 2

LITERATURE REVIEW

2.1 EXISTING SYSTEM WITH SECURITY APROACHES

Existing systems use IDS, firewalls, IPS, and SIEM to monitor network traffic and detect suspicious activities. Traditional IDS mainly uses predefined signatures, which may have limitations in detecting new or modified attacks. Machine learning-based IDS uses algorithms such as Decision Tree, Random Forest, and Gradient Boosting to classify network traffic. However, most systems mainly focus on attack detection and provide limited information about attack progression and suitable security responses.

Table 2.1 Literature Survey

S.NO	RESEARCH PAPER	TOOLS AND TECHNIQUES USED	APPLICATIONS USED	ADVANTAGES AND DISADVANTAGES	DATASET USED	INFERENCE FROM PAPER
1	A Generative and Explainable AI Framework for Real-Time Cyber Attack Prediction and Defense, Revathi M,Mallika P,Neeli Madhavi G .V,Pideka kundil Abhilash,Sri Lalitha IEEE, 2026.	Generative AI, XAI, Real-time prediction	Cyberattack prediction and defense	**Adv:** Prediction with explainability **Disadv:** Recommendations require validation; computationally demanding	Cybersecurity event data	Supports attack prediction and explainable defense
2	Intrusion Detection System with Explainable AI and Recommendation System, Soundariya S,Ramakrishnan ; Hari Pradap	ML, XAI, Recommendation system	IDS with security recommendations	**Adv:** Combines detection, explanation and recommendation **Disadv:** Recommendation depends on detection	Network intrusion data	Closely supports the proposed IDS recommendation system

	V,Gayathri M,varsha Shree P, IEEE, 2026.			accuracy and security rules		
3	AI-Driven Intrusion Detection and Automated Response for Encrypted Network Traffic,Aayushmaan Rajeev Bakshi,Rajalakshmi A,Ganesh Kumar S,Umamaheshwari J ,IEEE, 2026.	AI-based IDS, Anomaly detection, Automated response	Real-time intrusion detection and response	**Adv:** Faster response and reduced false positives **Disadv:** False positives may cause service disruption	Intrusion/security-event data	Supports automated or analyst-approved response
4	AI-Driven Real-Time Intrusion Detection and Automated Mitigation for Industrial IOT Process Control Sensors, Moulishwaran A,Visumathi J, IEEE, 2026.	Optimized Random Forest, Automated response	Real-time network IDS	**Adv:** 98.93% accuracy; 98.40% precision; 29% faster response **Disadv:** Limited modeling of long attack sequences and attacker intention	Network intrusion data	Supports Random Forest for detection and risk scoring
5	Machine Learning-Driven Threat Detection and Response Framework for Modern Cybersecurity Systems, Sathesh P. Sivashanmugam, IEEE, 2026.	ML, Threat detection, Mitigation framework	Network threat detection and response	**Adv:** Links attack classification with mitigation **Disadv:** Static policies may not adapt to new attacks	CICIDS2017	Supports attack-to-action mapping
6	Enhancing Intrusion Detection	AI-based IDS,	Reliable network attack detection	**Adv:** Focuses on reliability **Disadv:**	Network intrusion data	Supports evaluation of detection rate,

	Reliability Through Explanaible Machine Learning Models, Sakshi Khare,Simar Preet Singh,Vimal Kumar, IEEE, 2026.	Reliability analysis		Concept drift and new attack types require further evaluation		FPR and robustness
7	An Experimental Methodology to Predict Cyber Attacks Using Learning-Assisted Generative AI Technology, Deepa G,Geetha K,IEEE, 2026.	Generative AI, Learning-assisted prediction	Cyber-threat detection and attack prediction	**Adv:** Supports AI-based attack prediction experiments **Disadv:** Predictions require validation against real attack sequences	Cybersecurity attack scenarios	Supports experimental attack prediction
8	Hybrid Ensemble-based AI Intrusion Detection Framework Leveraging Sequential Deep Learning and Feature-Optimized Machine Learning for Real-Time Multi-Tier Network Threat Classification ,Ajeeth A,Sivanesan A,Boddu Prabhas,Sagar Varaprasad N V,Avaneesh R,Sri Sabarish R, IEEE, 2026.	Hybrid ensemble learning, Threat classification, Human verification	Intrusion detection and automated response	**Adv:** Fast response with human verification **Disadv:** Multiple models increase complexity and inference time	Network intrusion data	Supports human-in-the-loop recommendations

9	AI-Based Intelligent Network Intrusion Detection and Traffic Analyzer, Parimala S,Harhavardhan J S,Nicale J,Darshan Shivaji,Srimuthy ananjayan K,IEEE, 2026.	ML traffic analysis, Intelligent IDS	Network traffic analysis and intrusion detection	**Adv:** Combines traffic analysis with AI detection **Disadv:** Does not forecast attacks or recommend responses	Network traffic data	Supports traffic analysis and feature extraction
10	A Network Intrusion Detection System Using Machine Learning Techniques,Sayantika Maji,Khushburani,Nitin Siva Srinivas G,Saroj Kumar Panigrahy, IEEE, 2025.	ML classifiers, Comparative evaluation	Network intrusion classification	**Adv:** Compares multiple detection algorithms using UNSW-NB15 **Disadv**: Focuses mainly on classification	UNSW-NB15	Supports baseline model comparison
11	Artificial Intelligence Driven Intrusion Detection for SDN Environments,Samiha Sislam,Shahran Rahman Alve,Muhammad Zawad Mahmud,Faija	AI/ML classifier, SDN	SDN intrusion detection	**Adv:** Supports programmable and centralized networks Disadv: SDN architecture may require adaptation	UNSW-NB15	Supports automated flow-rule recommendations

	islam Oishe,Mohammad Monirujjaman Khan, IEEE, 2025.					
12	Intrusion Detection in IoT Networks Using LSTM Deep Learning Models with the UNSW-NB15 Dataset, Nazek Hassouneh,Saleh Al-sharaeh, IEEE, 2025.	LSTM deep learning	IoT intrusion detection	**Adv:** Captures temporal and sequential patterns **Disadv:** Computationally expensive	UNSW-NB15	Supports next-attack prediction
13	Enhancing Intrusion Detection Systems with Synthetic Attack Data and Advanced Classification Models , Mohyminul Islam,Ali Turab,Muhammad Ibrar,Younus Khan,Farhan Ullah,Umar Raza IEEE, 2025.	Synthetic data, ML/DL	Handling class imbalance in intrusion detection	**Adv:** Improves minority attack detection **Disadv:** Synthetic data may not represent real attacks	NSL-KDD, UNSW-NB15, CICIDS2017	Supports data augmentation for rare attacks
14	AI-Driven Cybersecurity Threat Detection: A Hybrid CNN-LSTM Deep Learning Approach,Haijia Zhong,Fucai Xie,Lanke Yu,Xianpeng Wang,Qi Xie,Mey Sok, IEEE, 2026.	Hybrid CNN, Deep learning	Cybersecurity threat detection	**Adv:** 89.20% F1-score and 97.33% attack recall on UNSW-NB15 **Disadv:** Requires more resources; limited action explanation	UNSW-NB15	Supports CNN-based feature learning

15	FocalCA: A Hybrid-Convolutional-Attention Encoder for Intrusion Detection on UNSW-NB15 Achieving High Accuracy Without Data Balancing, Amin Amiri ,Alireza Ghaffarnia,Shah newaz Karim Sakib,Dalei Wu,Yu Liang, IEEE, 2025.	Hybrid convolution, Attention encoder	Cyberattack detection	**Adv:** Highlights important traffic features Disadv: Difficult to deploy on resource-limited devices	UNSW-NB15	Supports feature identification and explanation
16	An Ensemble Learning Approach for Cyber Attack Detection on NSL-KDD and UNSW-NB15 Datasets , Hatice candan,Mustafa Yasin Erten,Huseyin Aydilek, IEEE, 2025.	Ensemble learning, Multiple classifiers	Network intrusion detection	**Adv:** Improves generalization **Disadv:** Increased complexity and difficult explanations	UNSW-NB15	Supports ensemble-based detection
17	Intrusion Detection for Wireless Sensor Network Using Particle swarm optimization Based Explainable Ensemble Machine Learning Approach, Shaikh Afnan	Practical Swarm Optimization, Ensemble ML, SHAP, LIME	Wireless sensor-network intrusion detection	**Adv:** Feature optimization and explanations; 99.65% peak accuracy **Disadv:** Increased training time; limited generalization	NSL-KDD, UNSW-NB15, CICIDS2017	Supports feature selection and SHAP/LIME explanations

	Birahim , Avijit Paul, Fahmida Rahman , Yamina Islam, Tonmoy Roy Mohammad Asif Hasan, Fariha Haque IEEE, 2025.					
18	Improved Coati Optimization Algorithm for Feature Selection Enabled Intrusion Detection in Internet of Things, Laith H. Alzubaidi , Rekha Cm ,Kavitha Mv, Ravi Kiran Varma P, V. Thirumurugan ,IEEE, 2025.	Coati Optimization, Random Forest	IoT intrusion detection	**Adv:** Reduces irrelevant features; 99.55% accuracy **Disadv:** Computational overhead; dataset-dependent	UNSW-NB15	Supports optimized feature selection
19	Enhancing Machine Learning Model Interpretability in Intrusion Detection Systems through SHAP Explanations and LLM-Generated Descriptions , Abderrazak Khediri , Hamda Slimi , Ayoub Yahiaoui , Makhlouf Derdour,Hakim Bendjenna,Charaf Eddine Ghenai, IEEE, 2024.	SHAP, LLM	Human-readable IDS explanations	**Adv:** Converts model results into understandable explanations **Disadv:** LLM explanations require validation	CICIDS2017	Supports natural-language attack explanations

20	Research on Network Security Threat Detection and Defense Mechanism Based on Artificial Intelligence, Yunmin Ding , IEEE, 2024.	Deep Learning, Q-learning	Threat detection and defense optimizatio	**Adv:** Connects detection with adaptive defense **Disadv:** Requires suitable reward function and safe environment	Network traffic and malware data	Supports adaptive security recommendations
21	Enhancing Autonomous Intrusion Detection System with Generative Adversarial Networks, Kevin Kostage,David West, Tim Meinert, Chengyi Qu,Prasad Calyam, Luca Mazzola ,IEEE, 2024.	Generative Adversarial Network; autonomous IDS	Intrusion detection with imbalanced data	**Adv:** Improves minority attack detection **Disadv:** Generated samples may be unrealistic	Imbalanced intrusion datasets	Supports augmentation for rare attacks
22	Unraveling Network Attacks with Machine Learning and Explainable AI, Deepak Kumar K,Saravana Gokul G, Anandhi S,Senthil Pandi S ,IEEE,.2024	ML, Explanaible AI	Network attack classification and interpretation	**Adv:** Improves understanding of model decisions **Disadv:** Explanation does not guarantee correct response	Network intrusion data	Supports feature-importance visualization

23	Evaluating Large Language Models for Enhanced Intrusion Detection in Internet of Things Networks, Ebelechukwu Nwafor, Ujjwal Baskota,Md Salik Parwez,Jeremy Blackstone,Habeeb Olufowob ,IEEE, 2024.	LLM, Ensemble IDS	IoT threat detection and analysi	**Adv:** Supports security-event interpretation **Disadv:** LLMs may be costly, slow and inconsistent	Real-world IoT intrusion datasets	Supports optional LLM-based security reports
24	All Predict Cost Efficient Decides:A New Cost-Centric Ensemble Learning Method for Network Intrusions Detection, Zhiyan Chen,Murat Simsek, Poonam Lohan,Burak Kantarci, Petar Djukic ,IEEE GLOBECOM, 2024.	Cost-sensitive ensemble learning	Cost-aware network intrusion detection	**Adv:** Considers attack and response costs **Disadv:** Requires reliable cost estimates	Network intrusion data	Supports risk-based recommendations
25	Enhancing Intrusion Detection Systems Using Machine	Decision Tree, Random Forest, Naïve Bayes,	Modern network and IoT intrusion detection	**Adv:** Compares multiple classifiers **Disadv:**	CSE-CIC-IDS2018	Supports dataset and baseline classifier evaluation

	Learning Classifiers on CSE-CIC-IDS2018 Dataset, Khalil Ibrahimi,Mohammed Jouhari,Zineb Jakout , IEEE, 2024.	Gradient Boosting		Benchmark may not represent modern changing traffic		

2.2 RESEARCH GAPS

The reviewed studies demonstrate that machine learning and deep learning techniques have been increasingly applied to network intrusion detection, cyberattack prediction, explainable security analysis, and automated response. Several existing approaches focus on improving attack classification accuracy using machine learning, ensemble learning, deep learning, and feature optimization techniques. Other studies have explored explainable artificial intelligence, generative AI, and intelligent security response. However, the reviewed approaches generally focus on individual stages of cybersecurity analysis rather than providing a unified framework that connects attack detection with attack progression analysis, risk assessment, and security recommendations.

A further gap is observed in the analysis of the relationship between successive attack stages. Conventional intrusion detection systems primarily identify whether network traffic is benign or malicious and, in some cases, classify the specific attack type. Such classification provides information about the current detected activity but does not necessarily indicate how an attack may progress to a subsequent stage. The proposed system addresses this aspect by incorporating MITRE ATT&CK-based attack analysis and a data-driven attack-chain prediction mechanism to analyse possible subsequent attack stages.

The reviewed approaches also indicate the need for practical and interpretable security decision support. Attack classification alone may not provide sufficient information for a security analyst to understand the attack context or determine appropriate actions. Similarly, approaches that depend heavily on complex AI or generative models may introduce additional computational requirements. Therefore, the proposed system aims to combine machine-learning-based attack classification with MITRE ATT&CK analysis, attack-chain prediction,

risk assessment, and security recommendations in a single framework. The major research gaps identified from the reviewed literature and their corresponding requirements in the proposed system are summarized in Table 2.2

Table 2.2 Research Gaps

S.NO	RESEARCH GAPS	REQUIREMENT IN PROPOSED SYSTEM
1	Limited Automated Security Recommendations	The proposed system plans to generate security recommendations based on detected attacks, attack-chain information, and risk.
2	Limited Detection of Rare Attacks	The proposed system should use suitable network-flow features for reliable attack classification.
3	Lack of Attack Explainability and Response	The proposed system aims to extend attack classification with threat analysis and security recommendations.
4	High Cost and Inconsistency of LLM-Based IDS	A machine-learning-based approach can provide an efficient initial attack classification before advanced security analysis.
5	Limited Attack-to-Attack Relationship Analysis	The proposed system includes attack-chain prediction to analyse possible subsequent attack stages.
6	Need for an Integrated Security Analysis Framework	The proposed system aims to integrate attack classification, MITRE ATT&CK analysis, attack-chain prediction, risk assessment, and security recommendations.
7	Limited Risk-Based Decision Support	The proposed system aims to incorporate attack information and risk assessment for security decision support.

2.3 PROPOSED SYSTEM

This suggested methodology is a Network Intrusion Detection and Security Recommendation System using machine learning techniques along with threat intelligence for detecting and analysing the attacks carried out on the network. The dataset used in the process is CICIDS2017, and the data is preprocessed to be able to perform classification. Random Forest classifier is implemented for the classification of network traffic between benign traffic and other attack classes. Once the attack is identified, MITRE ATT&CK tactics and techniques are mapped from the identified attacks to give further insights about the behaviour of the attacker. Based on the identified attacks and its context, security recommendations are provided.

In this proposed methodology, all of these aspects like data preprocessing, machine learning based network intrusion detection, attack classification, MITRE ATT&CK mapping, and security recommendations are combined together in one single process. This entire process can be evaluated using metrics for classification like accuracy, precision, recall, F1 score, and confusion matrix.

2.3.1 OVERVIEW

The following steps can be followed to detect and analyze network threats using the proposed method. The first step involves gathering of network traffic data from the CICIDS2017 dataset. Data cleaning and preprocessing techniques are applied on the gathered data, which include handling missing values and eliminating unwanted data to extract necessary features.

The data is then split into training and testing datasets. The model is trained using the training data and then the Random Forest classifier is used to categorize the testing network traffic as benign or attacks. Next, the attack is categorized into MITRE ATT&CK tactic and technique. Lastly, security recommendations are generated based on the attack detected and its threat context.

2.3.2 OBJECTIVES

The key objectives of the proposed AI-Driven Attack Chain Prediction and Security Recommendation System include:

- Collecting and analysing network traffic data comprising normal and abnormal activities in order to learn about different attack patterns.
- Preprocessing and organising network traffic data through dealing with missing values, filtering out irrelevant data and preparing features for the use of machine learning.

- Identifying significant features of network flow that could capture the characteristics of different cyberattacks.
- Developing an AI-enabled prediction model of attack that would be able to learn about patterns within network traffic and detect different types of attacks.
- Classifying the detected network activities into different attack types such as DDoS, DoS, Port Scanning, Brute Force, Botnet, Web Attacks, and Infiltration.
- Mapping the discovered attack activities to the corresponding MITRE ATT&CK tactics and techniques in order to learn about the behaviour and objectives of the attacker.
- Analysing the attack chain sequence and discovering the relationship between different attack phases.
- To predict the likely future attack techniques/stages from the detected attack behavior and threat intelligence.
- To provide security recommendations using the predicted attack stage, MITRE ATT&CK context, and detected security risks.
- To facilitate security analysis in a proactive manner by offering information about future attack activities that may occur.
- To measure the effectiveness of the AI-based prediction model using metrics such as accuracy, precision, recall, F1-score, and confusion matrix.
- To design a framework of integrated security decision support system that will incorporate attack detection, attack chain prediction, MITRE ATT&CK mapping, and security recommendations.

2.3.3 ADVANTAGES

Advantages of the proposed system include the following:

- Automated Detection: Eliminates the requirement for continuous monitoring of network traffic manually.
- Classification Using Machine Learning: Random Forest will learn from network traffic and classify attack categories.
- Detection of Multi-Class Attacks: Detects more than one attack class instead of only focusing on one attack category.
- MITRE ATT&CK Contextual Information: Helps to understand the tactics used by attackers in their attacks.
- Security Advice: Offers appropriate security advice depending on the identified threat.

CHAPTER 3

SYSTEM DESIGN

3.1 SYSTEM ARCHITECTRURE

The proposed system uses CICIDS2017 network traffic data for attack detection and analysis. The data is first preprocessed to remove inconsistencies and prepare it for analysis. Relevant network-flow features are then selected for the classification process. The selected features are given to the Random Forest model to classify network attacks. The detected attacks are mapped to the MITRE ATT&CK framework for further analysis. The system then performs attack-chain prediction and generates security recommendations.

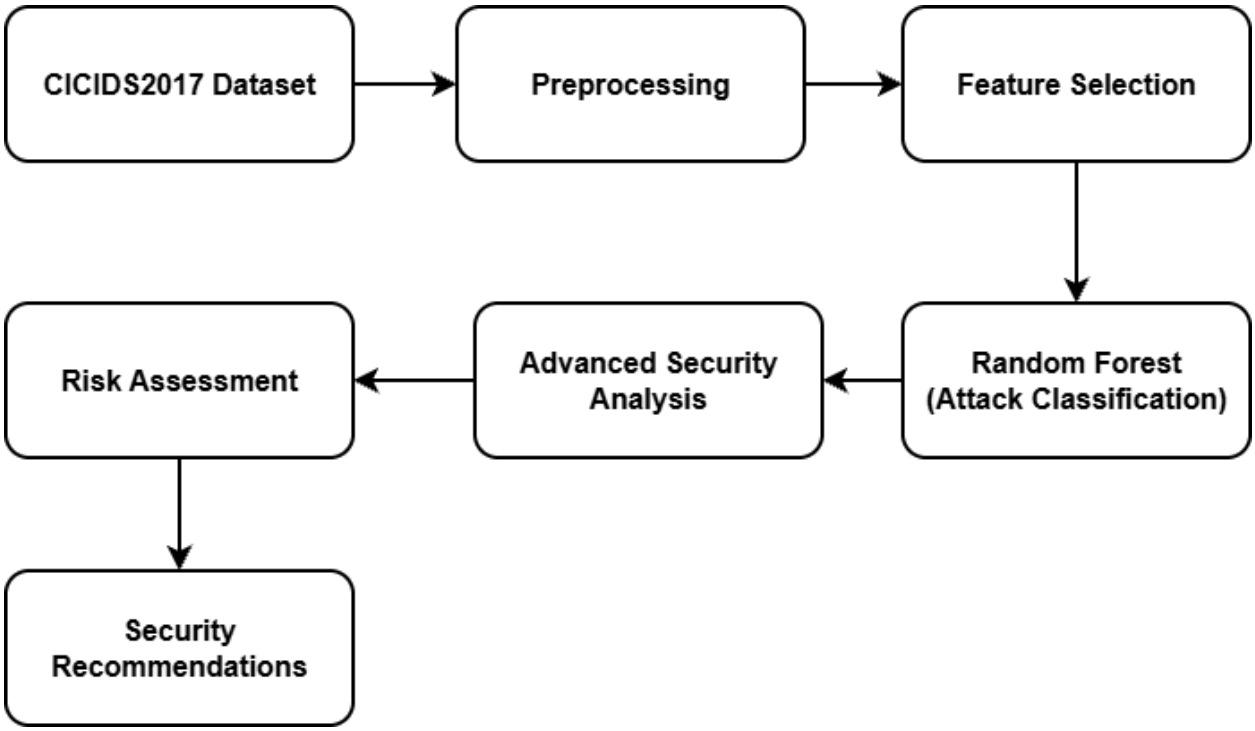


Figure 3.1 Architecture Diagram

3.2 MODULE DESIGN

This AI-Driven Attack Chain Prediction and Security Recommendation System is composed of five modules. Each of the modules is designed to carry out certain tasks so that the whole process results in the transformation of network traffic data into attack classification, threat analysis, attack chain prediction, and security recommendations.

Selection and Preprocessing of Datasets

This module is related to the selection and preprocessing of CICIDS2017 datasets of network traffic. Datasets are cleaned, merged, and properly organized so that consistent sets of

records of network traffic are obtained. Preprocessing is done to prepare the data for feature selection and machine learning.

Feature Selection

This module identifies a compact set of flow features useful for attack classification purposes. These features characterize some important aspects of network traffic and can be easily extracted from network traffic data.

Random Forest Attack Classification

The module uses the Random Forest classifier to determine whether the traffic category is benign or malicious. Network flow features that were selected earlier are input to the model. The classification of various types of network attacks is done via training and evaluating the Random Forest model on the prepared dataset.

Advanced Security Analysis and Threat Intelligence

In this module, an advanced analysis of the classified network traffic is conducted. It includes Zeek-based feature extraction, MITRE ATT&CK mapping, attack-chain prediction, and dynamic risk assessment. The classified attack is mapped against relevant MITRE ATT&CK tactics and techniques, whereas attack-chain prediction can be used to find out possible stages of an attack to follow. Dynamic risk assessment gives additional risk details about the detected attack.

Intelligent Security Recommendation

This module generates security recommendations based on the detected attack type, MITRE ATT&CK context, attack chain prediction, and risk information. Recommendations give security guidance for further analysing and handling the detected threat.

3.3 DATA FLOW DIAGRAM

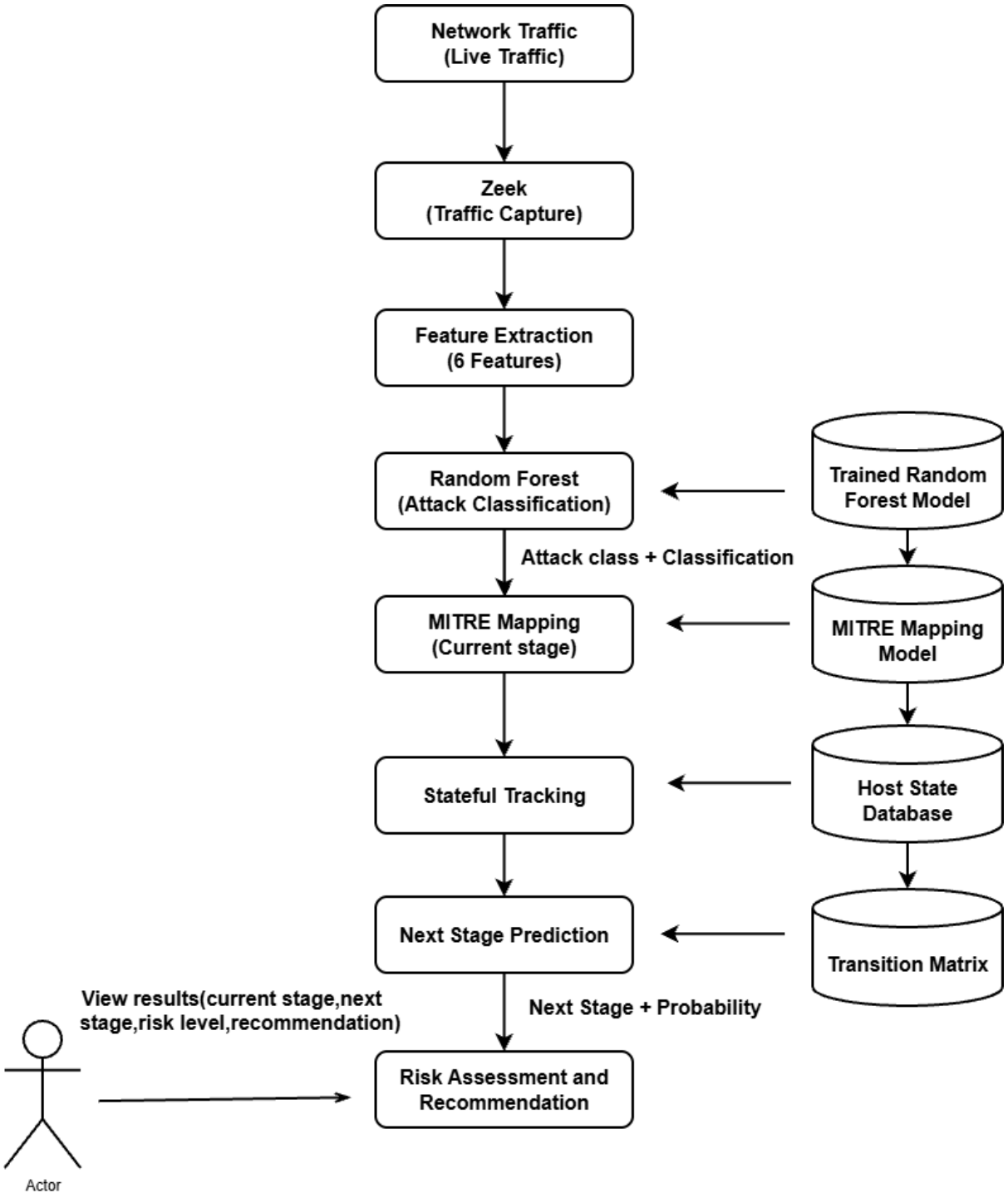


Figure 4.2 Dataflow Diagram

3.4 THREAT MODEL AND ATTACK SURFACE

This is because the AI-Driven Attack Chain Prediction and Security Recommendation System analyzes the network traffic data, machine learning features, attack classifications, threat intelligence, and security recommendations. Hence, various parts of the system are vulnerable to attacks from adversaries. The threat model describes the major assets, potential threats, attack surfaces, and security measures for the proposed system.

Assets

The major assets for the proposed AI-Driven Attack Chain Prediction and Security Recommendation System include the CICIDS2017 network traffic data set, selected network flow features, Random Forest model, attack classification data, MITRE ATT&CK mapping data, attack chain prediction data, dynamic risk assessment data, and security recommendations.

Potential Threats

- Manipulation of Data: Network traffic data or dataset entry could be manipulated for the sake of altering the analysis and classification process.
- Poisoning the Data: Malicious data that gets introduced to the training data set may manipulate the machine learning process and cause wrong predictions.
- Manipulation of the Model: Manipulation of the trained model of Random Forest may alter the performance of classification and cause the wrong attack to occur.
- Adversarial Input: Carefully manipulated network traffic data might try to bypass the machine learning model.
- Unauthorized Access: Unregistered users may try to get unauthorized access to the dataset, trained models, prediction results, and security recommendations.
- Manipulation of Recommendations: The introduction of wrong attack or risk information may lead to wrong security recommendations.
- Denial of Service: Too much data may affect the system's availability and processing capability.

Attack Surfaces

The important attack surfaces are the network traffic input, the dataset, feature processing step, the Random Forest model, MITRE ATT&CK mapping, attack-chain prediction, risk assessment, and security recommendation steps.

The network traffic input may have the manipulated or malicious data. The dataset may be the target for the manipulation or poisoning. The feature processing and Random Forest models

may be impacted by manipulated input or the unauthorized manipulation. MITRE ATT&CK mapping and attack-chain prediction steps may provide the wrong result on the basis of the wrong input information.

3.5 SECURITY CONTROLS

The AI-Driven Attack Chain Prediction and Security Recommendation System that is being designed needs proper security measures to protect the data set, machine learning model, threat analysis, and recommendations created by the system. The CICIDS2017 data set and preprocessed network traffic need to be protected from any form of unauthorized access or manipulation. Data that will be used in the model needs to be validated to minimize the risks of inaccurate or malicious data being entered into the system. Users who will have access to datasets, trained models, configurations, and security outcomes must be authorized users. The data set, features chosen, models' files, predictions, and configurations must be protected from any form of tampering.

The trained Random Forest model must be secured against any unauthorized alterations or modifications. Traffic data and features need to be validated before being fed into the model. The key system activities, attacks, attack chain predictions, risks, and recommendations must be logged. The MITRE ATT&CK mappings and threat information need to be validated in order to make sure that we have accurate attacks and predictions. The recommendations must be made using the attack, predicted attack chain, and risks. Recommendations need to be considered as decision support only and must be validated before implementing any security actions. These measures help in securing our proposed system.

CHAPTER 4

SYSTEM DESCRIPTION

4.1 DATA COLLECTION

AI-Driven Attack Chain Prediction and Security Recommendation System will make use of CICIDS2017 dataset as the main source of network traffic data. This data set contains both normal and malicious network traffic data associated with various types of cyberattacks. This data set contains network-flow data which can be used for training and testing the machine-learning based attack classification model. This dataset comprises traffic data gathered on different days and has different attack types, including DDoS, DoS, Port Scanning, Brute Force, Botnet, Web Attacks, and Infiltration. Traffic data record comprises many network-flow features and a label associated with the respective class, whether it is a normal traffic or any type of attack.

This combined data is further used for organizing the collected dataset into a standard format by extracting necessary traffic data from the collected dataset files. Network flow characteristics along with their respective labels are kept intact for further processing. The processed data is further fed into the Data Preprocessing stage for handling missing values, inconsistent data entries, and extraneous information prior to feature selection and machine learning. The prepared network traffic data is further utilized as an input for the Feature Selection stage and finally for training and testing the Random Forest-Based Attack Classification stage. Further, the output of the attack classification stage is fed into the Advanced Security Analysis and Threat Intelligence stage for carrying out MITRE ATT&CK Mapping, Attack Chains prediction, and Dynamic Risk Assessment.

4.2 DATA PREPROCESSING

The data preprocessing stage prepares the CICIDS2017 network traffic data for machine learning-based attack classification. The preprocessing process includes combining the required dataset files, selecting the relevant network-flow features, converting the feature values into numerical format, handling invalid and infinite values, preparing the target labels, analysing the class distribution, and dividing the dataset into training and testing sets.

The CICIDS2017 dataset contains network traffic records representing both benign and malicious activities. The required CSV files are first combined into a single dataset. After combining the files, the required columns are selected for further processing. Six network-flow features are selected as input features: Destination Port, Flow Duration, Total Fwd Packets,

Total Backward Packets, Flow Bytes/s, and Flow Packets/s. The Label column is used as the target variable.

The input feature set is represented as:

$$X = \{x_1, x_2, x_3, x_4, x_5, x_6\} \quad (4.1)$$

where,

$$X = \begin{Bmatrix} \text{Destination Port, Flow Duration, Total Fwd Packets, Total Backward Packets,} \\ \text{Flow Bytes/s, Flow Packets/s} \end{Bmatrix}$$

The target variable is represented as:

$$Y = \text{Label}$$

Therefore, the classification problem can be represented as:

$$f(X) \rightarrow Y$$

where f represents the machine learning classification function.

Numerical Conversion

The selected network-flow features are converted into numerical values to ensure that the data is in a suitable format for machine learning. Values that cannot be converted into valid numerical values are treated as invalid values. The numerical conversion is represented as:

$$X_i' = \begin{cases} \text{float}(X_i), & \text{if } X_i \text{ is a valid numerical value} \\ NaN, & \text{otherwise} \end{cases} \quad (4.2)$$

where X_i represents the original feature value, X_i' represents the converted value, and NaN represents an invalid or missing value.

Handling Infinite Values

Network-flow calculations may result in positive or negative infinite values. These values are identified and converted into missing values before further processing. The operation is represented as:

$$X_i' = \begin{cases} NaN, & X_i = +\infty \text{ or } X_i = -\infty \\ X_i, & \text{otherwise} \end{cases} \quad (4.3)$$

This prevents invalid infinite values from being provided as input to the machine learning model.

Removal of Invalid Records

After identifying missing and invalid values, records containing invalid values in the selected features are removed from the dataset. The cleaned dataset can be represented as:

$$D_{\text{clean}} = \{r_i \in D \mid r_i \text{ contains no invalid values}\} \quad (4.4)$$

where D represents the original dataset, D_{clean} represents the cleaned dataset, and r_i represents an individual network traffic record.

The target labels are also checked and prepared so that each valid network-flow record is associated with a corresponding traffic class. This ensures that the input features and target labels remain correctly aligned during model training and testing.

Class Distribution Analysis

The CICIDS2017 dataset contains different numbers of records for different traffic classes. Therefore, the class distribution is analysed during preprocessing. The number of records belonging to a particular class $c$is calculated using:

$$N_c = \sum_{i=1}^{N} \mathbb{I}\,(Y_i = c) \qquad (4.5)$$

where $N_c$represents the number of records belonging to class c, Nrepresents the total number of records, $Y_i$represents the class label of the i^{th}record, and $\mathbb{I}$represents the indicator function. The percentage of records belonging to a particular class is calculated as:

$$P_c = \frac{N_c}{N} \times 100 \qquad (4.6)$$

where $P_c$represents the percentage of records belonging to class c. This analysis helps to identify the distribution of benign and malicious traffic in the dataset.

Train-Test Split

After preprocessing, the cleaned dataset is separated into input features and target labels. The dataset is then divided into training and testing subsets using an 80:20 ratio. The training data is used for training the Random Forest classification model, while the testing data is used to evaluate the trained model.

The train-test division is represented as:

$$D = D_{\text{train}} \cup D_{\text{test}} \qquad (4.7)$$

where D_{train}represents the training dataset and D_{test}represents the testing dataset.

The size of the training and testing datasets is calculated as:

$$| D_{\text{train}} | = 0.80 \, | D |$$

$$| D_{\text{test}} | = 0.20 \, | D |$$

For the prepared CICIDS2017 dataset, the testing set contains 514,528 records, representing approximately 20% of the complete dataset.

The resulting preprocessed dataset contains the selected six network-flow features and the corresponding traffic labels. The cleaned and divided dataset is then provided as input to the Random Forest classification model for attack classification.

Overall, the data preprocessing stage converts the raw CICIDS2017 network traffic data into a clean, consistent, and machine-learning-ready format. The preprocessing process ensures that invalid values are handled, relevant features are selected, traffic classes are identified, and the dataset is properly divided for model training and evaluation. This prepared dataset forms the foundation for the Random Forest-based attack classification implemented in the next stage.

4.2.1 METHODOLOGY

The pre-processing process is conducted in order to transform the selected CICIDS2017 network traffic data into a clean and consistent format suitable for machine learning classification. In this process, the required CSV files are loaded and combined to form a unified dataset consisting of records representing both benign and malicious traffic. The selected data is then cleaned and prepared for training the Random Forest classification model.

The six selected network-flow features include Destination Port, Flow Duration, Total Fwd Packets, Total Backward Packets, Flow Bytes, and Flow Packets. The Label column is used as the target variable representing the traffic class. The following activities are performed during pre-processing:

- **Dataset Loading:** The required CICIDS2017 CSV files are loaded into the working environment.
- **Dataset Combination:** The required traffic records from the available CSV files are combined into a single dataset for further processing.
- **Feature Identification:** The six selected network-flow features are identified and selected as the input features for the classification model.
- **Numerical Conversion:** The selected feature values are converted into a suitable numerical format. Invalid values that cannot be converted into numerical values are identified.
- **Handling Invalid Values:** Infinite values and invalid numerical values are identified and handled before model training. Records containing invalid values are removed to obtain a clean dataset.
- **Target Label Preparation:** The Label column is isolated from the input features and is used as the target variable for attack classification.
- **Class Distribution Analysis:** The distribution of the different traffic classes is analysed to understand the number of benign and malicious records available in the dataset.
- **Feature and Target Label Isolation:** The preprocessed dataset is separated into two parts: the input feature data and the target label data. The input data contains the six

selected network-flow features, while the target data contains the corresponding traffic classes.

- **Training and Testing Split:** The cleaned dataset is divided into training and testing sets using an 80:20 ratio. The training data is used to train the Random Forest model, while the testing data is used for performance evaluation.
- The preprocessed data obtained after these steps is used as input for the Random Forest-based attack classification model.

4.2.2 RESULTS AND DISCUSSION

The implementation of the proposed system was carried out using the CICIDS2017 dataset. The dataset was first loaded and the required CSV files were combined into a single dataset for further processing. This provided a common structure for handling the network traffic records used in the machine learning model.

Figure 4.1 shows the loading and combination of the CICIDS2017 dataset. The combined dataset contains the network traffic records collected from the selected dataset files. This step provides the input data required for the subsequent preprocessing and classification stages.

```
================================================
CICIDS2017 DATASET LOADING AND COMBINATION
================================================
Number of CSV files loaded : 8
Total Records              : 3119345
Total Columns              : 85

First 5 Records:
```

	Flow ID	Source IP	Source Port	Destination IP	Destination Port	Protocol	Timestamp	Flow Duration	Total Fwd Packets	Total Backward Packets	...	min_seg_size_forward	Active Mean
0	192.168.10.12-192.168.10.50-60148-22-6	192.168.10.12	60148.0	192.168.10.50	22.0	6.0	6/7/2017 1:00	188.0	1.0	1.0	...	32.0	0.000
1	192.168.10.12-192.168.10.50-60148-22-6	192.168.10.50	22.0	192.168.10.12	60148.0	6.0	6/7/2017 1:00	83.0	1.0	2.0	...	32.0	0.000
2	192.168.10.12-91.189.89.199-123-123-17	192.168.10.12	123.0	91.189.89.199	123.0	17.0	6/7/2017 1:00	99947.0	1.0	1.0	...	40.0	0.000
3	192.168.10.12-72.38.129.202-123-123-17	192.168.10.12	123.0	72.38.129.202	123.0	17.0	6/7/2017 1:00	37017.0	1.0	1.0	...	32.0	0.000
4	8.0.6.4-8.6.0.1-0-0-0	8.6.0.1	0.0	8.0.6.4	0.0	0.0	6/7/2017 1:00	111161336.0	147.0	0.0	...	0.0	1753752.625

5 rows × 85 columns

FIGURE 4.1. LOADING AND COMBINING CICIDS2017 DATASET

The required network-flow attributes were then selected from the combined dataset. Six features, namely Destination Port, Flow Duration, Total Fwd Packets, Total Backward Packets, Flow Bytes/s, and Flow Packets/s, were used as input features. The Label column was considered as the target variable representing the corresponding traffic class.

Figure 4.2 shows the selected network-flow features and the target label used for the Random Forest classification model. These features represent different characteristics of network communication, including destination information, connection duration, packet volume, and traffic rate.

```
...  ============================================================
     SELECTED FEATURES AND TARGET LABEL
     ============================================================

     Input Features:
     1. Destination Port
     2. Flow Duration
     3. Total Fwd Packets
     4. Total Backward Packets
     5. Flow Bytes/s
     6. Flow Packets/s

     Target Label:
     Label

     Input Feature Shape: (3119345, 6)
     Target Label Shape : (3119345,)
```

	Destination Port	Flow Duration	Total Fwd Packets	Total Backward Packets	Flow Bytes/s	Flow Packets/s
0	22.0	166.0	1.0	1.0	0.000000	12048.192770
1	60148.0	83.0	1.0	2.0	0.000000	36144.578310
2	123.0	99947.0	1.0	1.0	960.509070	20.010606
3	123.0	37017.0	1.0	1.0	2593.403031	54.029230
4	0.0	111161336.0	147.0	0.0	0.000000	1.322402

FIGURE 4.2. SELECTED NETWORK-FLOW FEATURES AND TARGET LABEL

The selected data was then preprocessed before model training. Invalid numerical values and infinite values were handled to maintain consistency in the input data. The cleaned features were separated from the target labels, and the dataset was divided into training and testing data using an 80:20 ratio. The training data was used to train the model, while the testing data was kept for evaluating the model on unseen records.

The preprocessing stage ensures that the Random Forest model receives valid and consistent numerical input for the training process. The cleaned dataset is divided into training and testing sets using an 80:20 ratio. The training dataset is used to train the model, while the testing dataset is used to evaluate its performance on unseen records.

After preprocessing, the Random Forest classification algorithm is trained using the training dataset. Random Forest uses multiple decision trees to learn patterns between the network-flow features and their corresponding traffic classes. After training, the model predicts the classes of the testing records.

The performance of the Random Forest model is evaluated using accuracy, precision, recall, F1-score, classification report, and confusion matrix. These evaluation measures help determine how effectively the model classifies different network traffic categories. The results

obtained from the implementation demonstrate the feasibility of using the selected network-flow features with a Random Forest classifier for network attack classification. The preprocessing and train-test split provide consistent data for model training and enable evaluation using previously unseen data.

Table 4.1 Classification Report

Label	Precision	Recall	F1-Score	Support
BENIGN	1.00	1.00	1.00	454265
Bot	0.60	0.61	0.60	391
DDOS	1.00	1.00	1.00	25605
DoS GoldenEye	0.97	0.97	0.97	2059
DoS Hulk	0.99	0.99	0.99	46025
DoS Slowhttptest	0.98	0.98	0.98	1100
DoS slowloris	0.99	0.99	0.99	1159
FTP-Patator	1.00	1.00	1.00	1587
Heartbleed	1.00	1.00	1.00	2
Infiltration	0.80	0.57	0.67	7
PortScan	0.99	0.99	0.99	31761
SSH-Patator	0.98	0.99	0.98	1180
Web Attack-Brute Force	0.24	0.24	0.24	301
Web Attack-Sql Injection	1.00	0.25	0.40	4
Web Attack-XSS	0.06	0.06	0.06	130
Accuracy			1.00	565576
Macro avg	0.84	0.77	0.79	565576
Weighted avg	1.00	1.00	1.00	565576

CHAPTER 6
CONCLUSION

The AI-Driven Attack Chain Prediction and Security Recommendation System was developed to improve network attack detection and security analysis using machine learning techniques. The CICIDS2017 dataset was selected as the network traffic dataset, and the required CSV files were combined and preprocessed to obtain a consistent dataset for classification.

Six network-flow features, namely Destination Port, Flow Duration, Total Fwd Packets, Total Backward Packets, Flow Bytes/s, and Flow Packets/s, were selected as input features, while the Label column was used as the target variable. Invalid and infinite values were handled during preprocessing, and the dataset was divided into training and testing sets using an 80:20 ratio.

A Random Forest classification model was trained using the preprocessed network traffic data to classify different types of network activities. The model was evaluated using accuracy, precision, recall, F1-score, classification report, and confusion matrix. The obtained results demonstrate the feasibility of using the selected network-flow features for machine learning-based network attack classification.

The current implementation establishes the basic machine learning component of the proposed system. Further development will focus on MITRE ATT&CK mapping, data-driven attack-chain prediction, dynamic risk assessment, and security recommendation generation. These additional components can extend the system from basic attack classification toward a more comprehensive security analysis and decision-support framework.

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